INCH-POUND

MIL-C-24643/27D
22 November 1994
SUPERSEDING
MIL-C-24643/27C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LS2AU (INCLUDING VARIATION LS2A AND LS2AUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 22-7, maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Minimum average wall thickness 0.024 inch. Telephone identification code applied by method 3.
Fourth - Two conductors cabled together to form a pair, with nominal lay of 1-1/2 inches.
Fifth - Forty pairs cabled together with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.
Sixth - Binder tape, applied helically with overlap.
Seventh- Braided shield of tin coated copper. Minimum coverage of 85 percent.

Construction variant for type LS2AU

Eighth - Binder tape, applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

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Construction (non-watertight) (continued)

Construction variant type LS2A

Eighth - Binder tape, applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.
Tenth - Braided metal armor

Construction variant type LS2AUS

Eighth - Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Ninth - Braided shield of tin coated copper. Minimum coverage of 85 percent.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

TABLE I. Details

Military 1/ part no. M24643/27	Size 2/	Number of pairs	Diameter over Shielded core (inch nom.)	Overall diameter (inches)					
				TYPE LS2AU		TYPE LS2A		TYPE LS2AUS	
				min	max	min	max	min	max
-01U0	40	40	1.089	1.320	1.370	1.370	1.420	1.380	1.430

1/ Unarmored, overall shield. When armor option or double shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS2AU, LS2A and LS2AUS.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.71
Voltage withstand - volts, root mean square, minimum.	
Conductor to conductor	2500
Shield to shield (type LS2AUS only, overall shields only)	200
Conductor to shield	1250
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	5000
Shield to shield (type LS2AUS only, overall shields only)	100
Jacket flaws	No failure
Conductor and shield continuity	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Group A:	
Visual and dimensional	No failure
Capacitance - at 1 MHz, pF/FT maximum	
Mutual capacitance	20
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - at 0 ±2°C, mandrel diameter	
12 times cable diameter	No damage
Tear - pounds per inch thickness, minimum	
(ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - aged	
Insulation	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	5
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles minimum	125
Armor (type LS2A only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
* Surface transfer impedance (type LS2AUS only)	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60

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EXAMINATION AND TESTS (continued):	Requirements
Group D:	
Flame propagation (cable)	No failure
QUALIFICATION INSPECTION:	
Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
UNIT ORDERING LENGTH:	
1000 feet nominal.	

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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